

MATHEMATICAL ANALYSIS 1
HOMEWORK 3

- (1) Prove the following lemma:

Lemma. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be monotonically increasing on some $A \subseteq \mathbb{R}$. Then $f + g$ is also monotonically increasing on A . If either f or g are strictly increasing on A , then so is $f + g$. The same statements hold if we replace everywhere the word ‘increasing’ with the word ‘decreasing’.

- (2) Prove the following lemma:

Lemma. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$. Then:

$$\begin{array}{lll} f, g \text{ are both monotone increasing} & \Rightarrow & g \circ f \text{ is monotone increasing.} \\ f, g \text{ are both monotone decreasing} & \Rightarrow & g \circ f \text{ is monotone increasing.} \\ f, g \text{ are monotone of different kinds} & \Rightarrow & g \circ f \text{ is monotone decreasing.} \end{array}$$

- (3) Show that if $f : A \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is monotone increasing on A , then $-f$ is monotone decreasing on A .
- (4) Let t_c and s_c be the translation and scaling functions, respectively. Let r be the reflection function. Consider the function $f(x) = x^3$ restricted to the interval $[-1, 1]$. Sketch the following:
- $f \circ t_c$ for $c = -1, 0, 1$
 - $t_c \circ f$ for $c = -1, 0, 1$
 - $f \circ s_c$ for $c = \frac{1}{2}, 2$
 - $s_c \circ f$ for $c = \frac{1}{2}, 2$
 - The difference $f \circ r - r \circ f$.
- (5) Sketch the graphs of the following functions (choose a relevant part of the graph to sketch):
- $\sin(x), \sin(2x), \sin(-x)$
 - $\log_{\frac{1}{5}} x, \log_5 x, \log_5(x^2)$
 - $\cos(x), \cos(2x), \cos(x - \frac{\pi}{2})$
- (6) For the following pairs of functions f and g , compute $f \circ g$ and $g \circ f$, and determine their domains:
- $f(x) = x^2, g(x) = x + 1$
 - $f(x) = \sqrt{x}, g(x) = x - 2$
 - $f(x) = \frac{1}{x}, g(x) = x^2 + 1$
 - $f(x) = \sin(x), g(x) = 2x$
- (7) For the following functions f and subsets $A \subseteq \text{dom}(f)$, determine $\sup_A f$, $\inf_A f$, and whether the maximum and minimum are attained on A :
- $f(x) = x^2, A = [0, 2]$
 - $f(x) = x^2, A = (0, 2)$
 - $f(x) = x^3 + x^2 + 2x + 1, A = [0, 2]$
 - $f(x) = \frac{1}{x}, A = (0, 1]$
 - $f(x) = \frac{1}{x}, A = [1, +\infty)$
 - $f(x) = e^x, A = [-1, 1]$
- (8) Using trigonometric identities, simplify the following expressions:
- $\frac{\sin(x)}{\cos(x)} + \frac{\cos(x)}{\sin(x)}$
 - $(\sin(x) + \cos(x))^2 + (\sin(x) - \cos(x))^2$
 - $\frac{1 - \cos(2x)}{\sin(2x)}$
- (9) Prove the following equalities:
- $\tan(x) + \cot(x) = \frac{2}{\sin(2x)}$
 - $\cos(3x) = 4\cos^3(x) - 3\cos(x)$

Additional Practice Problems

★★ These problems are for you to practice ★★
★★ They are not part of the homework assignment ★★

- (1) Sketch the graphs of the following functions (choose a relevant part of the graph to sketch):
 - (a) $\sin(x + \frac{\pi}{2}), \sin(x) + 1$
 - (b) $\log_{10}(\frac{2}{x}), \ln(-x), \ln(x^{-1})$
 - (c) $-\cos(x), \cos(-x)$
 - (d) $\tan(x), \tan(2x), \tan(x + \frac{\pi}{4}), -\tan(x)$
 - (e) $\arcsin(x), \arcsin(2x), \arcsin(-x), \arcsin(x) + \frac{\pi}{2}$
 - (f) $\arctan(x), \arctan(2x), \arctan(-x), -\arctan(x)$
- (2) Sketch the graphs of the following power functions:
 - (a) $x^{1/2}, x^{1/3}, x^{1/4}$
 - (b) x^{-1}, x^{-2}, x^{-3}
 - (c) $x^{2/3}, x^{3/2}, x^{-1/2}$
 - (d) $(x-1)^{1/2}, (x+2)^{-1}, 2x^3$
- (3) For the following pairs of functions f and g , compute $f \circ g$ and $g \circ f$, and determine their domains:
 - (a) $f(x) = e^x, g(x) = \ln(x)$
 - (b) $f(x) = |x|, g(x) = x - 3$
 - (c) $f(x) = \sqrt{x}, g(x) = x^2$
 - (d) $f(x) = \frac{1}{x-1}, g(x) = \frac{1}{x}$
 - (e) $f(x) = x^{1/3}, g(x) = x^3$
 - (f) $f(x) = x^2 + 1, g(x) = \sqrt{x-1}$
- (4) For the following functions f and subsets $A \subseteq \text{dom}(f)$, determine $\sup_A f, \inf_A f$, and whether the maximum and minimum are attained on A :
 - (a) $f(x) = e^x, A = (-1, 1)$
 - (b) $f(x) = \sqrt{4-x^2}, A = (-2, 2)$
 - (c) $f(x) = \arctan(x), A = \mathbb{R}$
 - (d) $f(x) = \frac{1}{1-x}, A = [0, 1)$
 - (e) $f(x) = x(2-x), A = (0, 2)$
 - (f) $f(x) = x^{1/3}, A = [-8, 8]$
 - (g) $f(x) = x^{-2}, A = [1, +\infty)$
- (5) Determine whether the following functions are even, odd, or neither:
 - (a) $f(x) = x^4 + x^2$
 - (b) $f(x) = x^5 - x^3$
 - (c) $f(x) = x^3 + x^2$
 - (d) $f(x) = \frac{x}{x^2+1}$
 - (e) $f(x) = x^{-2}$
 - (f) $f(x) = |x| + x$
- (6) Using trigonometric identities, simplify the following expressions:
 - (a) $\sin(x+y)\sin(x-y)$
 - (b) $\cos^4(x) - \sin^4(x)$
 - (c) $\frac{1+\tan^2(x)}{1+\cot^2(x)}$
- (7) Prove the following equality:
 - (a) $\frac{\sin(x)+\sin(3x)}{\cos(x)+\cos(3x)} = \tan(2x)$
- (8) Simplify the following expressions involving powers:
 - (a) $(x^3)^{1/2} \cdot x^{-1/2}$
 - (b) $\frac{x^{2/3} \cdot x^{1/2}}{x^{-1/6}}$
 - (c) $(x^{-2}y^3)^{-1/2}$
 - (d) $\left(\frac{x^{1/3}}{x^{-2/3}}\right)^3$

HOMEWORK 3 SOLUTIONS

- (1) *Proof.* Let $x, y \in A$ with $x < y$. Since f and g are monotonically increasing:

$$f(x) \leq f(y)$$

$$g(x) \leq g(y)$$

Adding: $(f + g)(x) = f(x) + g(x) \leq f(y) + g(y) = (f + g)(y)$.

If either is strictly increasing, the corresponding inequality is strict, so $f + g$ is strictly increasing.

The decreasing case follows similarly. $\square$

- (2) *Proof.* Let $x < y$.

Case 1: Both increasing

$$f(x) \leq f(y)$$

$$g(f(x)) \leq g(f(y)) \quad (\text{since } g \text{ increasing})$$

So $g \circ f$ is increasing.

Case 2: Both decreasing

$$f(x) \geq f(y)$$

$$g(f(x)) \leq g(f(y)) \quad (\text{since } g \text{ decreasing})$$

So $g \circ f$ is increasing.

Case 3: Different types

$$f(x) \leq f(y) \quad (\text{if } f \text{ increasing})$$

$$g(f(x)) \geq g(f(y)) \quad (\text{since } g \text{ decreasing})$$

So $g \circ f$ is decreasing. $\square$

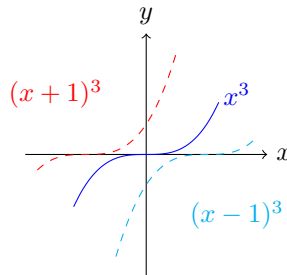
- (3) *Proof.* Let $x, y \in A$ with $x < y$. Since f is increasing:

$$f(x) \leq f(y) \Rightarrow -f(x) \geq -f(y)$$

So $-f$ is decreasing. $\square$

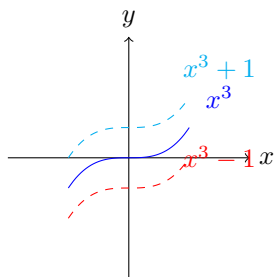
- (4) (a) $f \circ t_c$ for $c = -1, 0, 1$:

- $c = -1$: $f \circ t_{-1}(x) = f(x - 1) = (x - 1)^3$
- $c = 0$: $f \circ t_0(x) = f(x) = x^3$
- $c = 1$: $f \circ t_1(x) = f(x + 1) = (x + 1)^3$



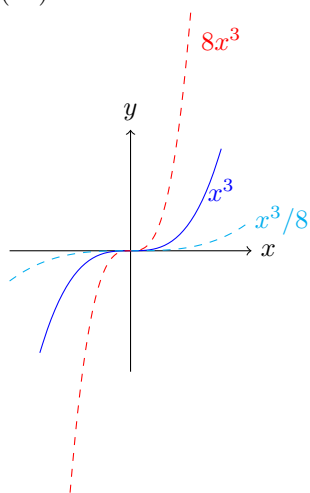
- (b) $t_c \circ f$ for $c = -1, 0, 1$:

- $c = -1$: $t_{-1} \circ f(x) = f(x) - 1 = x^3 - 1$
- $c = 0$: $t_0 \circ f(x) = f(x) = x^3$
- $c = 1$: $t_1 \circ f(x) = f(x) + 1 = x^3 + 1$



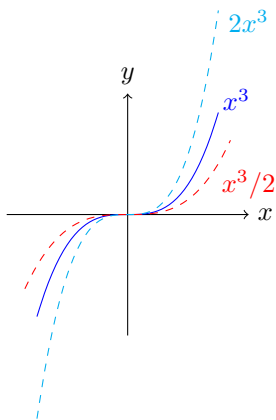
(c) $f \circ s_c$ for $c = \frac{1}{2}, 2$:

- $c = \frac{1}{2}$: $f \circ s_{1/2}(x) = f(x/2) = (x/2)^3 = x^3/8$
- $c = 2$: $f \circ s_2(x) = f(2x) = (2x)^3 = 8x^3$



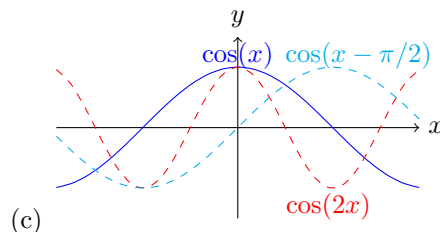
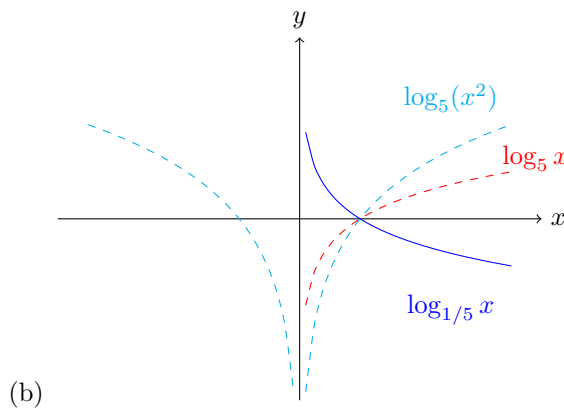
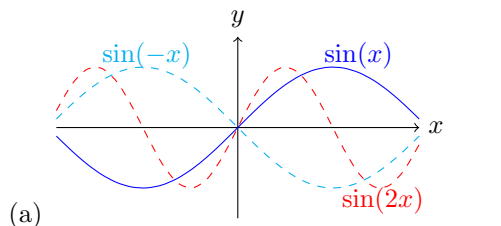
(d) $s_c \circ f$ for $c = \frac{1}{2}, 2$:

- $c = \frac{1}{2}$: $s_{1/2} \circ f(x) = \frac{1}{2}f(x) = \frac{1}{2}x^3 = x^3/2$
- $c = 2$: $s_2 \circ f(x) = 2f(x) = 2x^3$



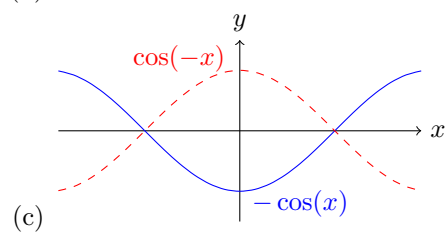
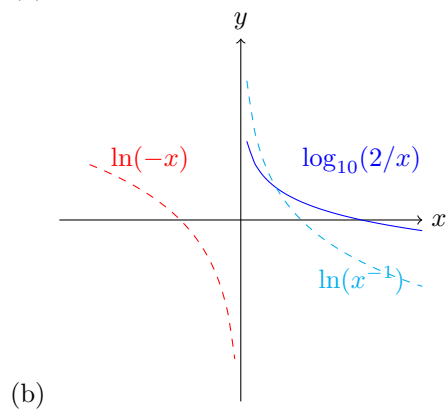
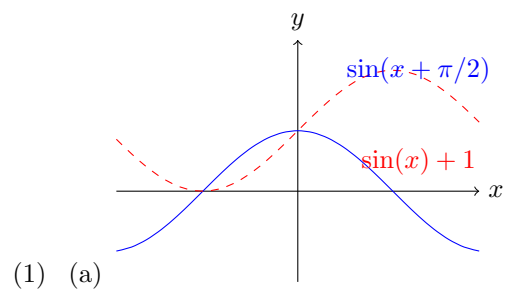
(e) $f \circ r - r \circ f = x^3 - (-x)^3 = x^3 + x^3 = 2x^3$

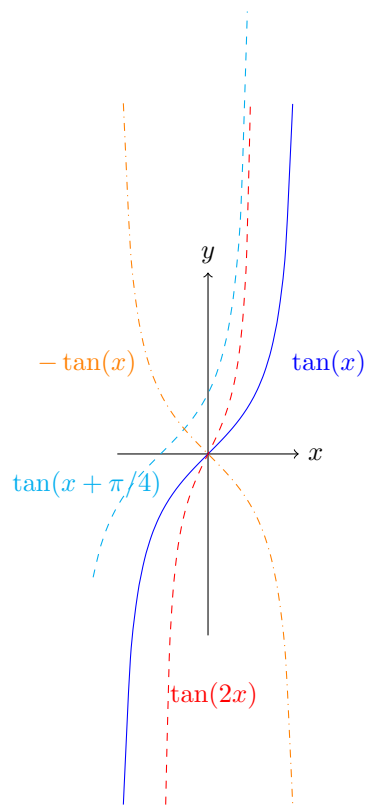
(5)



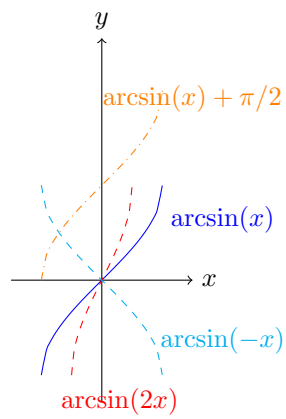
- (6) (a) $f \circ g(x) = (x+1)^2$, $\text{dom} = \mathbb{R}$
 $g \circ f(x) = x^2 + 1$, $\text{dom} = \mathbb{R}$
 (b) $f \circ g(x) = \sqrt{x-2}$, $\text{dom} = [2, \infty)$
 $g \circ f(x) = \sqrt{x} - 2$, $\text{dom} = [0, \infty)$
 (c) $f \circ g(x) = \frac{1}{x^2+1}$, $\text{dom} = \mathbb{R}$
 $g \circ f(x) = \frac{1}{x^2} + 1$, $\text{dom} = \mathbb{R} \setminus \{0\}$
 (d) $f \circ g(x) = \sin(2x)$, $\text{dom} = \mathbb{R}$
 $g \circ f(x) = 2 \sin(x)$, $\text{dom} = \mathbb{R}$
- (7) (a) $\sup = 4$, $\inf = 0$, max attained at $x = 2$, min at $x = 0$
 (b) $\sup = 4$, $\inf = 0$, neither attained
 (c) $\sup = 17$, $\inf = 1$, max at $x = 2$, min at $x = 0$
 (d) $\sup = \infty$, $\inf = 1$, min attained at $x = 1$, no max
 (e) $\sup = 1$, $\inf = 0$, max attained at $x = 1$, no min
 (f) $\sup = e$, $\inf = e^{-1}$, max at $x = 1$, min at $x = -1$
- (8) (a) $\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \tan x + \cot x = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \frac{1}{\sin x \cos x} = \frac{2}{\sin 2x}$
 (b) $(\sin x + \cos x)^2 + (\sin x - \cos x)^2 = (1 + 2 \sin x \cos x) + (1 - 2 \sin x \cos x) = 2$
 (c) $\frac{1 - \cos 2x}{\sin 2x} = \frac{2 \sin^2 x}{2 \sin x \cos x} = \frac{\sin x}{\cos x} = \tan x$
- (9) (a) $\tan x + \cot x = \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \frac{1}{\sin x \cos x} = \frac{2}{\sin 2x}$
 (b) $\cos 3x = \cos(2x + x) = \cos 2x \cos x - \sin 2x \sin x$
 $= (2 \cos^2 x - 1) \cos x - 2 \sin^2 x \cos x$
 $= 2 \cos^3 x - \cos x - 2(1 - \cos^2 x) \cos x$
 $= 2 \cos^3 x - \cos x - 2 \cos x + 2 \cos^3 x = 4 \cos^3 x - 3 \cos x$

ADDITIONAL PRACTICE PROBLEMS SOLUTIONS

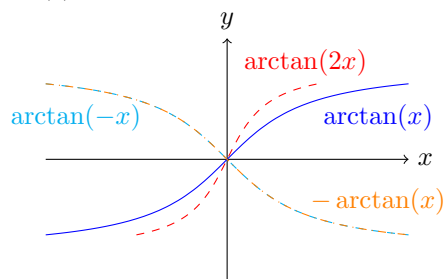




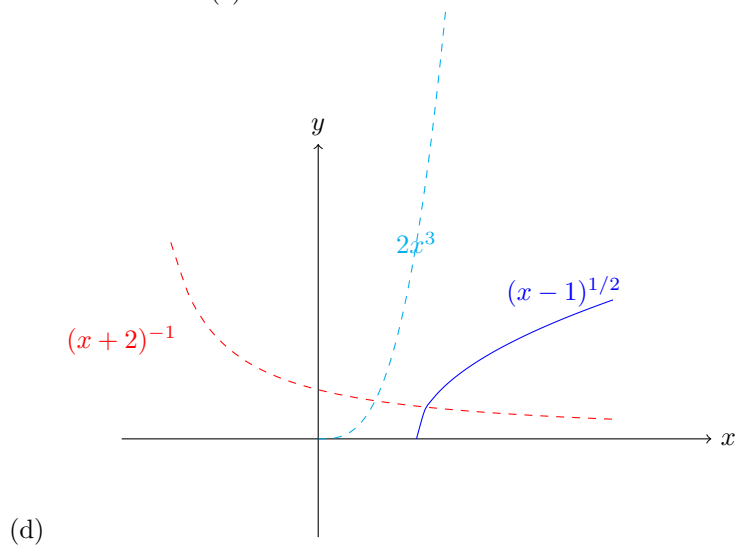
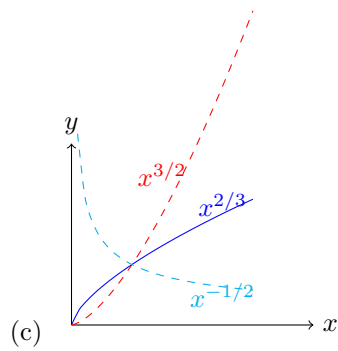
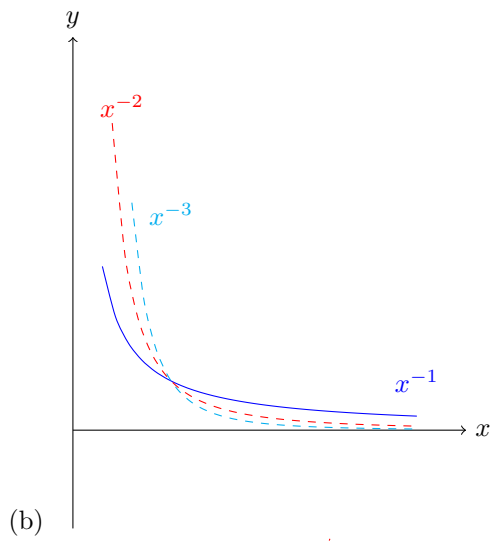
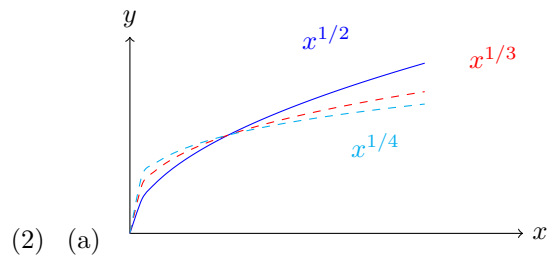
(d)



(e)



(f)



- (3) (a) $f \circ g(x) = e^{\ln x} = x$, $\text{dom} = (0, +\infty)$
 $g \circ f(x) = \ln(e^x) = x$, $\text{dom} = \mathbb{R}$
(b) $f \circ g(x) = |x - 3|$, $\text{dom} = \mathbb{R}$
 $g \circ f(x) = |x| - 3$, $\text{dom} = \mathbb{R}$
(c) $f \circ g(x) = \sqrt{x^2} = |x|$, $\text{dom} = \mathbb{R}$
 $g \circ f(x) = (\sqrt{x})^2 = x$, $\text{dom} = [0, +\infty)$
(d) $f \circ g(x) = \frac{1}{\frac{1}{x}-1} = \frac{x}{1-x}$, $\text{dom} = \mathbb{R} \setminus \{0, 1\}$
 $g \circ f(x) = \frac{1}{\frac{1}{x-1}} = x - 1$, $\text{dom} = \mathbb{R} \setminus \{1\}$
(e) $f \circ g(x) = (x^3)^{1/3} = x$, $\text{dom} = \mathbb{R}$
 $g \circ f(x) = (x^{1/3})^3 = x$, $\text{dom} = \mathbb{R}$
(f) $f \circ g(x) = (\sqrt{x-1})^2 + 1 = x$, $\text{dom} = [1, +\infty)$
 $g \circ f(x) = \sqrt{(x^2+1)-1} = |x|$, $\text{dom} = \mathbb{R}$
(4) (a) $\sup = e$, $\inf = e^{-1}$, neither attained
(b) $\sup = 2$, $\inf = 0$, neither attained
(c) $\sup = \pi/2$, $\inf = -\pi/2$, neither attained
(d) $\sup = \infty$, $\inf = 1$, min attained at $x = 0$, no max
(e) $\sup = 1$, $\inf = 0$, neither attained
(f) $\sup = 2$, $\inf = -2$, both attained
(g) $\sup = 1$, $\inf = 0$, max attained at $x = 1$, no min
(5) (a) Even: $f(-x) = (-x)^4 + (-x)^2 = x^4 + x^2 = f(x)$
(b) Odd: $f(-x) = (-x)^5 - (-x)^3 = -x^5 + x^3 = -f(x)$
(c) Neither: $f(-x) = (-x)^3 + (-x)^2 = -x^3 + x^2 \neq \pm f(x)$
(d) Odd: $f(-x) = \frac{-x}{(-x)^2+1} = \frac{-x}{x^2+1} = -f(x)$
(e) Even: $f(-x) = (-x)^{-2} = x^{-2} = f(x)$
(f) Neither: $f(-x) = |-x| + (-x) = |x| - x \neq \pm f(x)$
(6) (a) $\sin(x+y)\sin(x-y) = \frac{1}{2}[\cos(2y) - \cos(2x)] = \sin^2 y - \sin^2 x$
(b) $\cos^4 x - \sin^4 x = (\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x) = \cos 2x$
(c) $\frac{1+\tan^2 x}{1+\cot^2 x} = \frac{\sec^2 x}{\csc^2 x} = \frac{\sin^2 x}{\cos^2 x} = \tan^2 x$
(7) *Proof.* Using sum-to-product identities:

$$\begin{aligned} \frac{\sin x + \sin 3x}{\cos x + \cos 3x} &= \frac{2 \sin(2x) \cos x}{2 \cos(2x) \cos x} \\ &= \frac{\sin 2x}{\cos 2x} = \tan 2x \end{aligned}$$

□

- (8) (a) $(x^3)^{1/2} \cdot x^{-1/2} = x^{3/2} \cdot x^{-1/2} = x^1 = x$
(b) $\frac{x^{2/3} \cdot x^{1/2}}{x^{-1/6}} = x^{2/3+1/2-(-1/6)} = x^{2/3+1/2+1/6} = x^{4/3}$
(c) $(x^{-2}y^3)^{-1/2} = x^1y^{-3/2} = \frac{x}{y^{3/2}}$
(d) $\left(\frac{x^{1/3}}{x^{-2/3}}\right)^3 = (x^{1/3-(-2/3)})^3 = (x^1)^3 = x^3$